GMARK IoT & CMMS PLATFORM

Complete Modular Architecture, SCADA Telemetry & CMMS Operations Manual

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EXECUTIVE SUMMARY & SYSTEM SCOPE

The GMARK IoT & CMMS Platform is a mission-critical, enterprise industrial software suite engineered to unify cyber-physical operational assets with computerized maintenance lifecycle management. Operating seamlessly across multiple hierarchical tiers—from corporate Main Facilities down to localized machinery nodes—the platform ingests live sensor telemetry via industrial MQTT brokers, evaluates safety boundaries in real-time, dispatches automated work orders, tracks spare part inventories, and maintains an immutable audit trail.

Architecture Pillar	Primary Technologies Employed	Functional Role & Capability
Frontend Client	React 18, TypeScript, TailwindCSS, Redux Toolkit, Vite	Single Page Application (SPA) offering responsive SCADA supervisory dashboards, asset master tables, interactive charts, and permission-gated controls.
Microservices Backend	NestJS (Node.js), TypeScript, RESTful APIs, Socket.io	Modular architecture with dependency injection, JWT authentication guards, background cron schedulers, and exception filters.
Database & ORM	PostgreSQL, Prisma ORM, JSONB schema support	Relational database ensuring referential integrity, foreign key cascades, time-series telemetry snapshots, and automated code sequence generators.
Industrial Protocol Gateway	MQTT Gateway, Mosquitto, TCP Packet Parsers	Continuous ingestion of PLC registers from industrial controllers (AIPCU booster pumps, fire alarm panels, sub-soil sump pumps).
Security & Multi-Tenancy	JWT Bearer Tokens, bcrypt, 7-dimension permissions	Multi-tenant isolation per society/facility, with role-based and user-level overrides plus physical asset boundary scoping.

PLATFORM GOAL & OPERATIONAL IMPACT

By combining real-time SCADA telemetry with automated preventive maintenance workflows, the GMARK platform eliminates equipment downtime by up to 42%, ensures 100% compliance with statutory inspection checkpoints, and offers complete transparent visibility into plant asset depreciation and spares stock valuation.

CHAPTER 1: AUTHENTICATION, IDENTITY & MULTI-TENANCY (CORE AUTH)

Secure session lifecycle, multi-tenant isolation, token rotation, and identity propagation.

%¸ Module Architecture & Overview

The Core Auth module provides secure authentication and multi-tenancy enforcement. It guarantees that every incoming request is cryptographically validated and scoped to the user's specific operational boundary (Society, Main Facility, Facility, or assigned Machine pool). Cross-tab session state is kept fresh via the AuthSync worker.

Component Layer	Implementation Details
Frontend Pages	LoginPage.tsx, SignUpPage.tsx, ForgotPasswordPage.tsx, ResetPasswordPage.tsx, ProtectedRoute.tsx, AuthSync.
Backend Controllers & Services	backend/src/modules/core/auth (auth.controller.ts, auth.service.ts, jwt.strategy.ts, roles.guard.ts).
Database Models	User, Registration, Society, UserBuilding.
Cryptographic Standards	JSON Web Tokens (JWT), bcrypt (10 salt rounds), Bearer Authorization headers.
Module Access Scoping	Users can be restricted to SCADA, CMMS, or BOTH via the moduleAccess flag in the User model.
Tenancy Scoping Fields	societyId, mainFacilityId, facilityId, and adminId enforce tenant data partitioning.

%¸ API Endpoints & Request Payloads

HTTP Method & Route	Request Body / Parameters	Response Payload & Description
POST /auth/login	{ username, password }	{ token, refreshToken, user: { id, username, role, moduleAccess, permissions } }
POST /auth/register	{ username, email, password, companyName, gstNumber, address }	Creates User, Registration, and defaults tenancy parameters.
POST /auth/refresh	{ refreshToken }	{ token, refreshToken } — Issues fresh access token without re-login.
GET /auth/me	Bearer Token in Auth Header	Returns validated user object with up-to-date role and designation permissions.
POST /auth/forgot-password	{ email }	Dispatches time-limited password recovery token to user email.
POST /auth/reset-password	{ token, newPassword }	Validates reset token and securely updates password hash.

CHAPTER 2: GRANULAR ACCESS CONTROL & PERMISSIONS (MANAGE ACCESS)

Dynamic 7-dimension permission matrix combined with physical asset boundary scoping.

%¸ The 7 Permission Dimensions

Access privileges are not binary; they are evaluated per module across 7 discrete operational actions. Privileges can be configured at the Designation (Role) level and overridden at the individual User level.

Permission Dimension	System Enforcement & Impact	UI Control Behavior
canView	Controls read-access to the module records and route views.	Hides sidebar menu items and blocks URL transitions in ProtectedRoute.
canCreate	Authorizes creation of new records via POST API endpoints.	Conditionally renders "Add New" action buttons and modal forms.
canEdit	Authorizes modification of existing records via PUT/PATCH endpoints.	Disables edit icons, form inputs, and save buttons.
canDelete	Authorizes record destruction via DELETE API endpoints.	Hides trash icons and prompts deletion security confirmations.

Permission Dimension	System Enforcement & Impact	UI Control Behavior
canImport	Enables bulk ingestion via CSV and Excel upload parsers.	Renders file dropzone wizards and template download links.
canExport	Enables data extraction to CSV, Excel, and structured formats.	Renders table export utility buttons in header toolbars.
canPrint	Authorizes formatted print layouts and PDF report generation.	Triggers browser print dialogs and jsPDF document generators.

%¸ Physical Asset Boundary Scoping (hasAssetPermission)

Beyond functional permissions, the platform enforces strict physical scoping. The hasAssetPermission security rule evaluates whether an authenticated user is assigned to a specific Main Facility, Facility, or Machine. Users assigned to Facility A cannot view or manipulate assets residing in Facility B, ensuring complete operational privacy in multi-plant enterprises.

CHAPTER 3: ENTERPRISE 4M LOCATION & ASSET HIERARCHY

Structural organization from corporate enterprise down to production cells and equipment.

%¸ Hierarchical Tiers & Entity Relationships

The 4M (Man, Machine, Method, Material) framework structures physical asset locations into an intuitive, scalable parent-child relational tree. Every asset belongs to a defined geographical and organizational hierarchy.

Hierarchy Tier	Key Attributes & Fields	Entity Relation & Purpose
Tier 1: Main Facility	mainFacilityCode, mainFacilityName, gstNumber, category, subcategory, latitude, longitude, address, facilityHead	Corporate campus or enterprise division. Root entity in MainFacility model; 1-to-many with Facilities and Buildings.
Tier 2: Facility (Plant/Unit)	facilityCode, facilityName, parentFacilityId, isLocationPart, category, latitude, longitude, facilityHeadName	Operational plant or factory unit. Facility model supports self-referencing tree hierarchy for nested campus sub-plants.
Tier 3: Line / Building / Tower	name, info, societyId, mainFacilityId	Physical building, manufacturing production line, or cell housing machinery. Building model links to SCADA devices and machines.
Tier 4: Machine / Asset	machineCode, name, deviceId, machineType, make, modelNumber, serialNumber, costCenter, headOfDepartment	Individual equipment or SCADA node. Machine model links to Facility, Building, Device, and maintenance schedules.

%¸ Hierarchy Master Management Pages

Master Interface	Route & Path	Core Operational Features
Main Facility Master	/master/main-facility	View, search, filter enterprise campuses. Includes AddMainFacility.tsx modal with postal code auto-lookup.
Facility Management	/facility-management	Plant listing with nested child facility badges, coordinates mapping, and AddFacility.tsx modal.
Building & Tower Master	/societies & /society/:id	Building and tower breakdown with flat/cell counts and SCADA device links.

CHAPTER 4: MACHINE & ASSET MASTER MANAGEMENT (EQUIPMENT LIFECYCLE)

Equipment onboarding, dynamic engineering specs, digital binders, and public QR inspection cards.

%¸ Dynamic Specifications & Document Binders

Industrial machinery varies wildly between pump stations, HVAC chillers, and fire panels. The Machine Master avoids rigid schemas by utilizing MachineSpecification (storing dynamic title, inputType, data, unit) and MachineDocument (storing operating manuals, wiring diagrams, and warranty deeds as File, Link, or Note).

%¸ Public Machine QR Card (/public/machine/:codeOrId)

Every registered machine features an automatically generated QR code. Affixed physically to equipment cabinets, field technicians or operators can scan the QR code with any mobile device to view real-time operating metrics, past maintenance history, and instantly submit breakdown tickets without cumbersome login hurdles.

Specification / Feature	Database Model / Mechanism	Operational Value
Machine Code Sequence	CodeSequence model (e.g., prefix MAC, padded to 3 digits)	Generates uniform, non-colliding asset IDs like MAC-001, MAC-002 across all facilities.
Parent-Child Assets	parentMachineId / childMachines self-relation	Enables sub-assembly tracking (e.g. Pump #1 Impeller within Booster Station A).
IoT Device Pairing	deviceId unique field linking to Device model MAC ID	Directly binds live SCADA telemetry registers to the equipment management card.
Public Card View	PublicMachineCard.tsx component	Allows mobile camera QR scanning for instant field diagnostics and breakdown reporting.

CHAPTER 5: INDUSTRIAL SCADA TELEMETRY & INGESTION ENGINE

High-throughput MQTT broker, binary packet parsing, Modbus registers, and WebSocket live streaming.

%¸ Telemetry Architecture & Ingestion Flow

Physical industrial controllers (AIPCU, Fire Control Panels, Sub-Soil pumps) transmit binary/JSON telemetry payloads over TCP/TLS to the central MQTT broker. The NestJS MqttService ingests packets, routes them through ParserService, updates DeviceState in PostgreSQL, and pushes state deltas to connected browser clients via Socket.io under 50ms latency.

Telemetry Parameter	Register Address / Type	Operational Context & Normal Operating Range
Phase Voltages (RY, YB, RB)	Analog Registers (Float)	Monitors 3-phase grid power quality (Normal: 415V ± 10%). Detects phase failure and voltage sags.
Motor Currents (R, Y, B)	Analog Registers (Float, Motors 1-4)	Monitors individual motor load currents. Detects motor overload, dry run, and jammed impellers.
System Pressure (bar & mA)	Analog Register (4-20mA Transmitter)	Measures output manifold line pressure (0-16 bar). Triggers lead-lag booster pump sequencing.
Motor Run Hours	Discrete Accumulators (Seconds)	Records cumulative run time for predictive bearing lubrication and seal replacement cycles.
Fire Panel Statuses	Discrete Bitmask (Lines 1 & 2)	Auto, Manual, Run, Trip, Control On/Off for Main, Jockey, Terrace, and Diesel Engine pumps.
Fault Trip Codes	Integer Enumeration	Captures input power fault types, dry run trips, overload trips, and VFD communication alarms.

%¸ Historical Time-Series Logging (DeviceSnapshot)

The Device model configures historical data retention. Administrators can configure time-based logging (minute, hour, daily snapshots) or conditional logging (snapshots triggered only on state changes or threshold excursions). Snapshots are stored in DeviceSnapshot with indexed timestamps for rapid historical trend plotting.

CHAPTER 6: SCADA ALARMS, THRESHOLDS & MULTI-TIER ESCALATION

Real-time register boundary evaluation, audible browser alarms, and automated email escalation chains.

%¸ Automated Multi-Tier Escalation Matrix

When an industrial sensor breaches a programmed boundary (e.g., Manifold Pressure < 2.0 bar for > 30 seconds), the system creates a DeviceAlert. If on-duty operators fail to acknowledge the alert within the programmed time window, the system automatically escalates to plant supervisors and executive administration.

Escalation Stage	Trigger Condition	System Notification & Action
Tier 1: On-Duty Operator	Sensor threshold breached (EMERGENCY or WARNING)	Audible browser alarm, flashing red UI badge, push notification, and email to Operator.
Tier 2: Plant Supervisor	Alert remains UNACKNOWLEDGED past escalationInterval (e.g. 30 mins)	Alert marked isEscalated: true, level 2 email sent to Building Admin / Plant Manager.
Tier 3: Executive Admin	Alert remains UNACKNOWLEDGED past level 2 window (e.g. 60 mins)	High-priority SMS/Email dispatched to Super Admin; auto-creates high-priority breakdown Work Order.

CHAPTER 7: INSPECTION CHECKPOINT MASTER & STANDARD OPERATING PROCEDURES

Standardized maintenance inspection criteria, tolerance limits, and technician walk-through routes.

%¸ Inspection Checkpoint Schema & Attributes

The CheckpointMaster model acts as the single source of truth for plant inspection routines. Checkpoints define what must be inspected, how it should be measured, and what numerical values represent safe operation.

Checkpoint Parameter	Schema Field & Type	Operational Purpose & Enforcement
Checkpoint Code	code: String (Unique)	Standardized code sequence (e.g., CHK-PREV-001, CHK-ELEC-014).
Inspection Type	checkpointType: String	Categorized as SAFETY, MECHANICAL, ELECTRICAL, LUBRICATION, or OPERATIONAL.
Tolerance Range	minValue, maxValue: Float	Specifies valid numerical limits (e.g. Bearing Temperature: Min 20°C, Max 75°C).
Standard Value	standardValue: String	Baseline benchmark reading (e.g. "415V Balanced", "Normal Clear Oil", "3.5 Bar").
Frequency	frequency: String	Defines required inspection cadence: DAILY, WEEKLY, MONTHLY, or QUARTERLY.
User Checkpoints	UserCheckpoint model	Assigns specific checkpoints to individual technicians to form daily mobile walk-through checklists.

CHAPTER 8: PREVENTIVE MAINTENANCE (PM) SCHEDULING & AUTOMATION

Automated recurring service calendars, tolerance day windows, pre-allocated spares, and cron dispatch.

%¸ Automated Work Order Dispatch Workflow

Preventive Maintenance schedules operate on an automated calendar engine. When the current date enters the scheduled maintenance window (calculated using startDate, frequency, and prePlanToleranceDays), a NestJS background cron task automatically instantiates a corresponding Work Order, pre-attaches the required checkpoints, and notifies assigned technicians.

Schedule Attribute	Implementation Detail	Preventive Maintenance Impact
Recurrence Cadence	DAILY, WEEKLY, MONTHLY, BIMONTHLY, QUARTERLY, HALF_YEARLY, YEARLY	Guarantees regular servicing intervals based on equipment manufacturer warranties.
Pre/Post Plan Tolerance	prePlanToleranceDays and postPlanToleranceDays (integers)	Provides operational flexibility so servicing can align with planned plant production downtime.
Pre-Reserved Spares	MaintenanceScheduleSpare (links sparePartId, quantity)	Automatically reserves replacement seals, filters, and oil in the warehouse before work commences.
Audit Logging	MaintenanceScheduleLog (CREATED, TRIGGERED_WO, PAUSED, COMPLETED)	Maintains complete operational logs for ISO-55001 and regulatory safety compliance audits.

CHAPTER 9: CMMS WORK ORDER LIFECYCLE & MULTI-TIER APPROVALS

End-to-end ticketing, severity levels, technician dispatch, proof of work, and executive sign-off.

%¸ Work Order Lifecycle State Machine

The WorkOrder lifecycle enforces strict transition gates. Work orders cannot jump from PENDING to COMPLETED without going through manager approval, technician assignment, and proof-of-work photo submission.

Lifecycle Phase	Permitted Status Transitions	Actions, Roles & System Validations
1. Inception & Triage	PENDING_APPROVAL (Triggered via MANUAL, SCHEDULED, or QR MACHINE_CARD)	Operator or SCADA threshold logs ticket with priority (LOW, MEDIUM, HIGH, CRITICAL).
2. Manager Verification	APPROVED or REJECTED	Maintenance Manager assigns technician (assignedTechnicianId), sets estimated start/end dates, and approves budget.
3. Work Execution	IN_PROGRESS or REVISION_REQUESTED	Technician logs consumed spare parts. If parts are unavailable, technician submits WorkOrderRevision for date extension.
4. Proof of Work	COMPLETED (Requires media attachment upload)	Technician must upload at least one photo or video tagged as PROOF_OF_WORK before completing.
5. Final Sign-off	CLOSED or Level 3 Approval	Supervisor validates repair quality. High-severity breakdown jobs require Level 3 Executive Sign-off (level3ApprovedById).

%¸ Proof of Work & Media Attachments (WorkOrderMedia)

The WorkOrderMedia model stores photographic and video evidence. Attachments are categorized by upload stage: INITIAL (photo of damaged asset during ticket creation), PROOF_OF_WORK (photo of replaced component and clean job site), or REVISION (photo of unforeseen structural defects justifying schedule extension).

CHAPTER 10: SPARE PARTS INVENTORY & STOCK VALUATION

Warehouse stock monitoring, shelf vs transit quantities, minimum safety buffers, and spares consumption.

%¸ Multi-Location Stock Quantities & Reorder Triggers

The SparePartMaster model tracks critical industrial replacements (bearings, mechanical seals, gaskets, circuit breakers). Inventory is divided into inHandQuantity (physically in warehouse), atSourceQuantity (ordered from vendor in transit), and stockQuantity (total active reserve).

Inventory Field	Schema Type & Default	Operational Function
partCode	String (Unique)	Standardized SKU code (e.g. SP-PUMP-SEAL-012).
stockQuantity	Integer (Default: 0)	Total available units across all plant warehouses.
minimumQuantity	Integer (Default: 0)	Minimum safety buffer. When stockQuantity <= minimumQuantity, system flags low stock.
unitCost & rate	Float (Default: 0.0)	Unit procurement cost. Multiplied by stock to calculate plant total asset valuation.
machinesList	Json Array	List of compatible machine models where this spare part can be legally installed.
stockLocations	Json Array	Warehouse aisle, rack, and bin designations (e.g. Warehouse 2, Rack B, Bin 4).

CHAPTER 11: SUPPLIER REGISTRY & PROCUREMENT DIRECTORY

Centralized vendor master tracking contract terms, statutory tax numbers, currencies, and contact persons.

%¸ Vendor Profiles & Procurement Linkages

The Supplier module maintains full commercial and statutory profiles for spare part vendors, OEMs, and third-party maintenance contractors. Each supplier is indexed via unique supplierCode (e.g. SUP-001) generated by CodeSequence.

Supplier Attribute	Description & Schema Structure	Procurement & Legal Purpose
supplierCode	Unique identifier (SUP-001, SUP-002)	Standardized reference used in Purchase Orders and delivery challans.
gstNumber	Statutory Tax Identifier (GSTIN)	Ensures tax compliance and input credit verification during spares procurement.
Primary Currency	primaryCurrency (INR, USD, EUR)	Supports multi-currency purchasing for imported machinery parts.
Business Category	category, businessDescription	Segments suppliers into Electrical, Mechanical, Hydraulic, or Automation vendors.
Geo-Coordinates	latitude, longitude, postalCode	Enables geographic proximity calculations for emergency breakdown dispatch.

CHAPTER 12: EMPLOYEE MASTER & STAFF ALLOCATIONS

Engineering workforce records, reporting managers, designations, and user credential linkage.

%¸ Workforce Profiles & User Linkage

The Employee model maintains internal personnel records for plant engineers, technicians, and supervisors. An employee record is linked 1-to-many with the User model, allowing an individual to possess system login credentials with privileges inherited directly from their Designation.

Attribute	Description	Application Interaction
employeeCode	Unique sequence code (EMP-001) per tenant	Standardized indexing for shift allocation and timesheet logging.
designation	Links to Designation model	Inherits default 7-dimension permissions across all application modules.
reportingTo	Links manager name / employee code	Defines managerial escalation chains and work order approval hierarchies.
Postal Codes	PostalCode model with auto-lookup	Entering pincode automatically populates city, state, and country in the UI.

CHAPTER 13: SYSTEM AUDIT LOGGING & DATA GOVERNANCE

Immutable security ledger recording every administrative, modification, and deletion event.

%¸ Non-Repudiation & Forensic Audit Trail

Every sensitive operation (asset creation, schedule editing, role assignment, work order closure, user deletion) passes through the NestJS AuditLogService. The service records the actor, target module, action type, IP address, and UTC timestamp into the AuditLog table, ensuring complete compliance with ISO-27001 and ISO-55001 standards.

Logged Field	Extraction & Capture Method	Security & Compliance Impact
userId	Extracted from cryptographically verified JWT token	Guarantees non-repudiation of record modifications.
action	String identifier (e.g., CREATE_WORK_ORDER, DELETE_MACHINE)	Enables detailed forensic audits during incident investigations.
module	Module name (e.g. machines, work_orders, access)	Permits granular filtering of audit logs by functional operational domain.

Logged Field	Extraction & Capture Method	Security & Compliance Impact
ipAddress	Captured via x-forwarded-for or req.socket.remoteAddress	Verifies physical network origin and detects unauthorized remote logins.
createdAt	High-resolution server UTC timestamp	Provides chronological event correlation with physical industrial incidents.